SIMATS ENGINEERING
CO3 - ASSESSMENT TOOL - 1: Experiments

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

Name	**D.Raghava**
Register Number	**192525304**

COURSE OUTCOME COVEREDCO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications.* (BL3)
Assessment Tool Weightage: 40%

Dave's Taxonomy: Precision (Level 3) **Total Marks: 40**
Question Count: 4

Duration : 60 Minutes

Code of Conduct
I D.Raghava(192525304) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student: D.Raghava

Experiments Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Experimental Design	25%	Design is systematic and technically strong	Mostly correct design	Basic design with gaps	Poorly designed activity
Use of Tools / Platforms	20%	Appropriate and effective use of cloud tools	Good tool usage with minor issues	Limited tool usage	Incorrect or missing tool usage
Application of Concepts	20%	Concepts are applied accurately to practical tasks	Mostly accurate application	Partial application	Weak application
Analysis and Output	20%	Strong analysis and meaningful outcome interpretation	Good analysis with minor gaps	Basic analysis	Little or no analysis
Documentation	15%	Clear, structured, and complete documentation	Mostly complete documentation	Partially complete	Poor documentation

Total Marks Awarded:

Assessment Tasks

Experiments

Experiment 1: Create a simple cloud software application for Library book reservation system for SIMATS library using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Title of the book, author, publications, Year, edition, no of copies, Rack no, status (taken/returned) etc.

Experiment 2: Create a simple cloud software application for Payroll Processing System using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Name of the employee, experience in present org , over all experience, basic pay, HRA, DA, TA, net pay, gross pay etc.

Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 1 CPU, 2GB RAM and 15GB storage disk using a Type 2 Virtualization Software.

Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Cloning of a VM and Test it by loading the Previous Version/Cloned VM

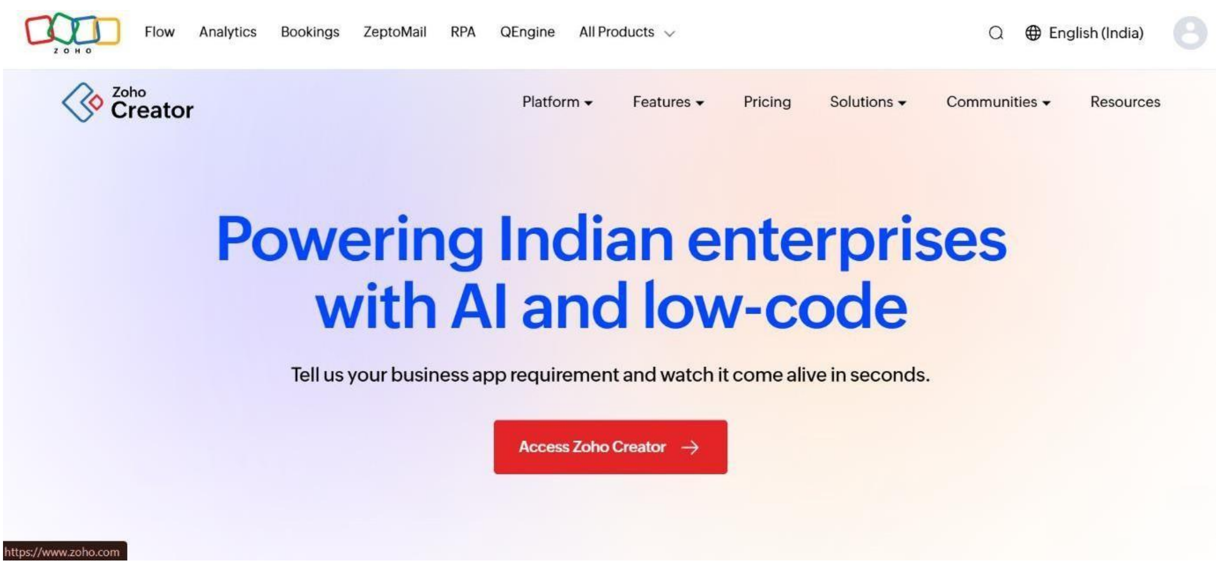
1) Create a simple cloud software application for Library book reservation system for SIMATS library using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Title of the book, author, publications, Year, edition, no of copies, Rack no, status (taken/returned) etc.

AIM:

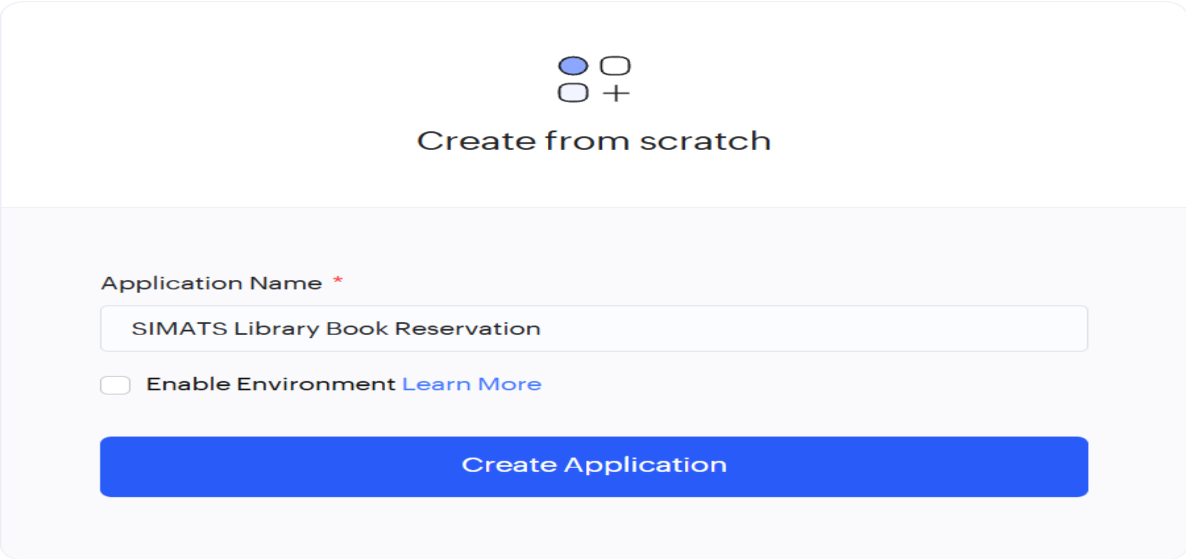
To create a simple cloud software application for Library Book Reservation System for SIMATS Library using any Cloud Service Provider to demonstrate SaaS.

PROCEDURE:

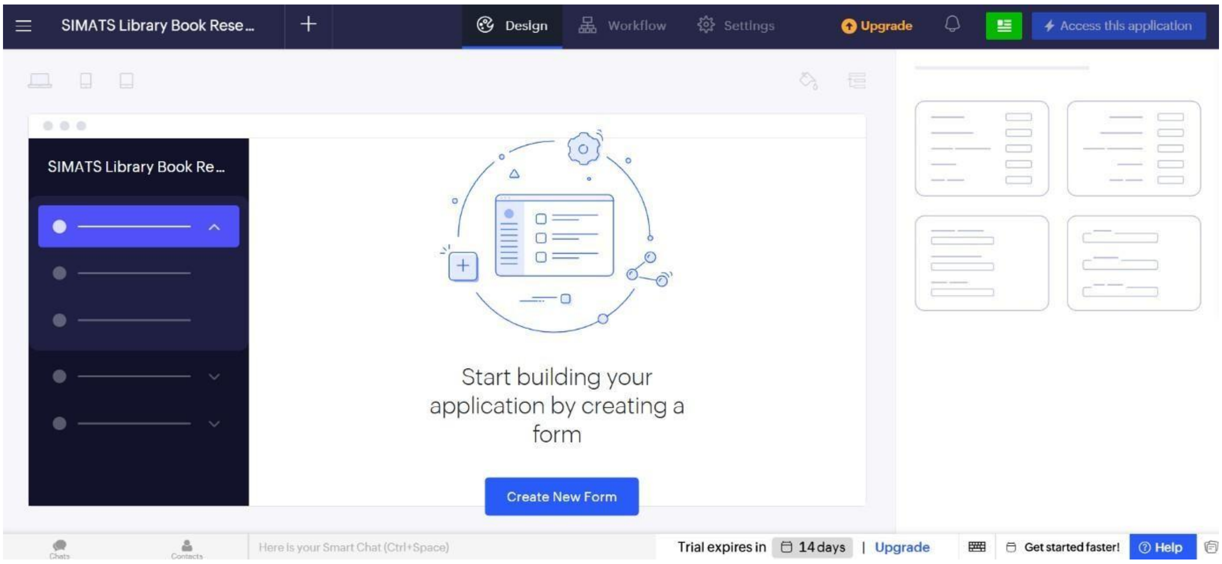
STEP 1: Log into zoho.com**.**

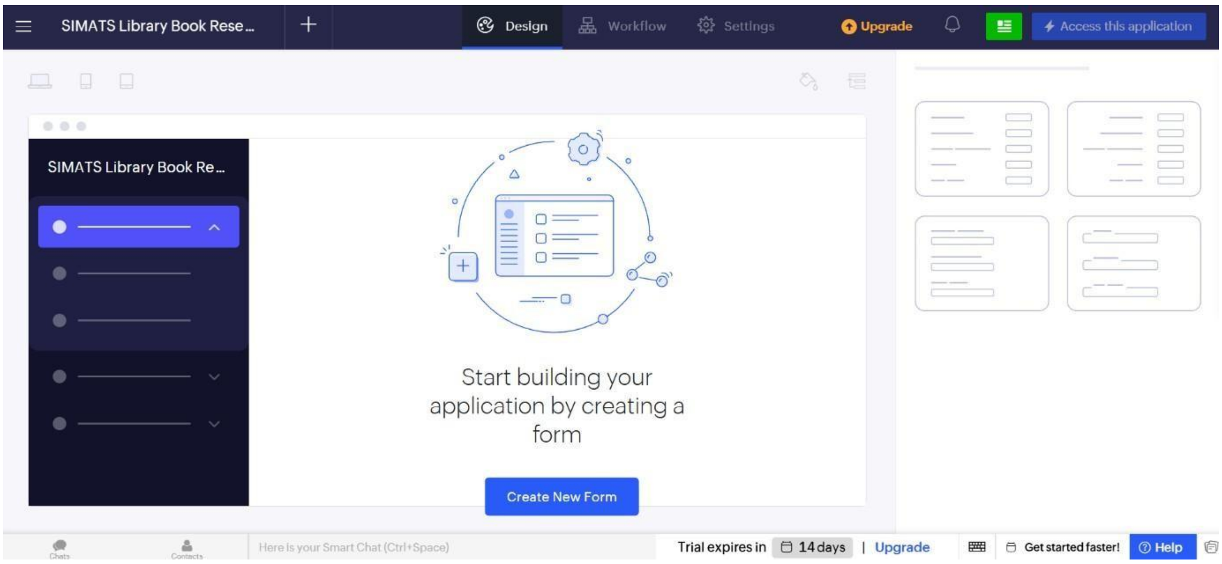


STEP 2: Create a new application from scratch in Zoho Creator.

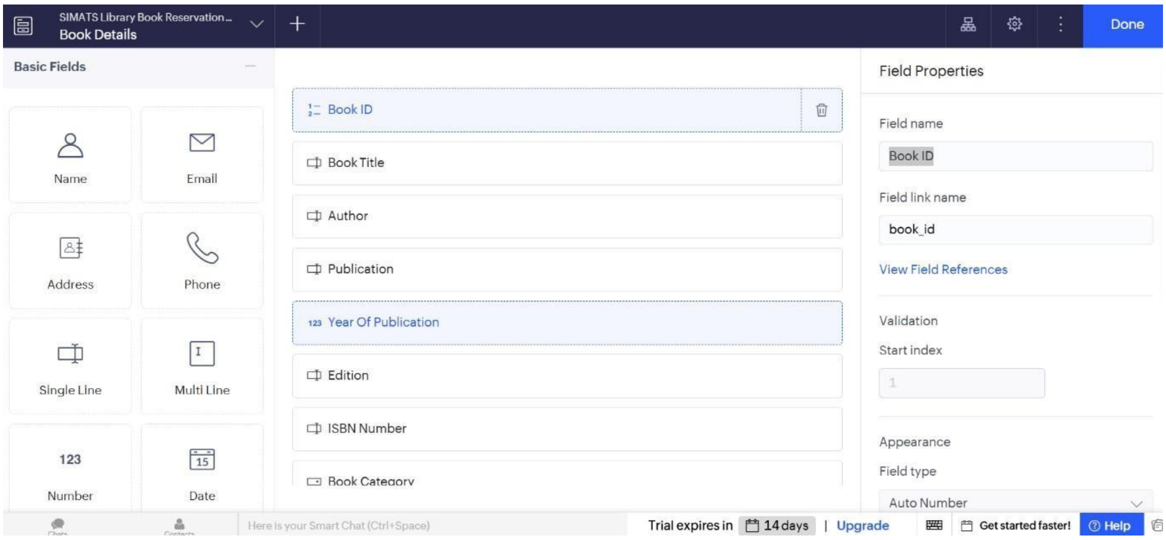


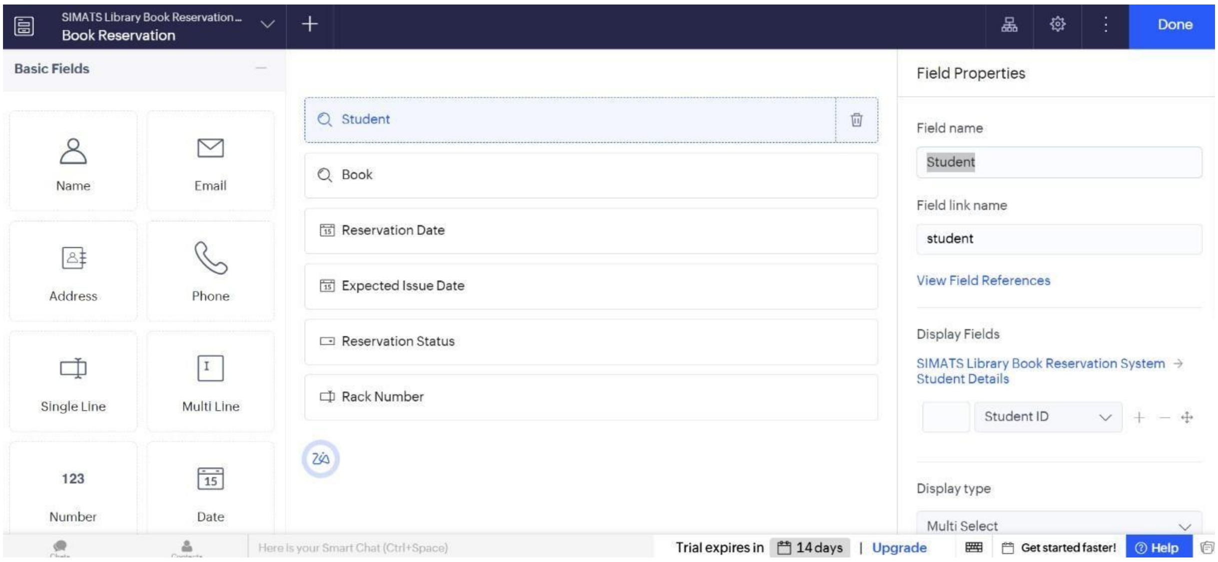
STEP 3: Create a new application and create forms from scratch.

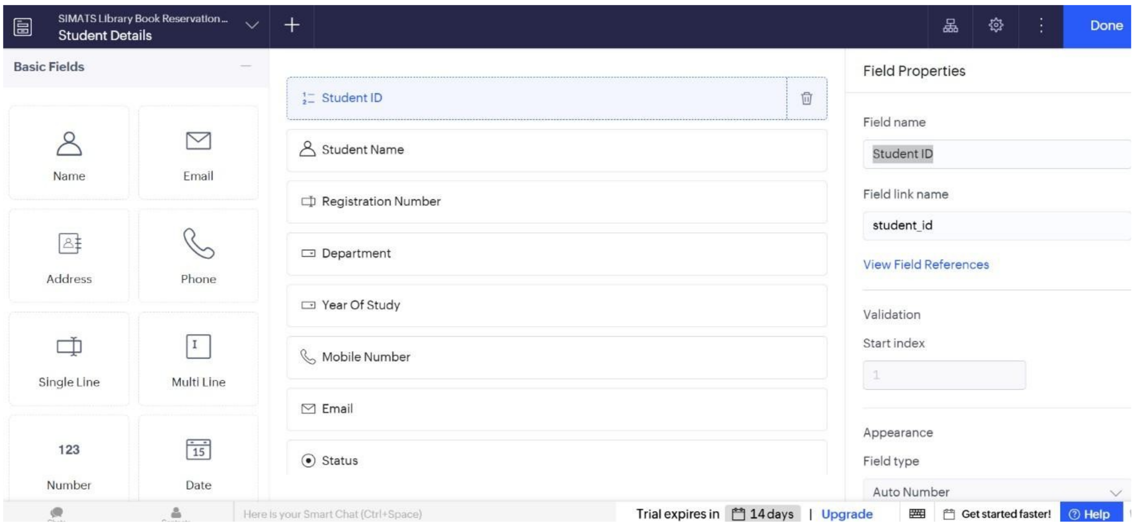




STEP 4: Create Book Details Form.







STEP 6: Create Book Reservation Form

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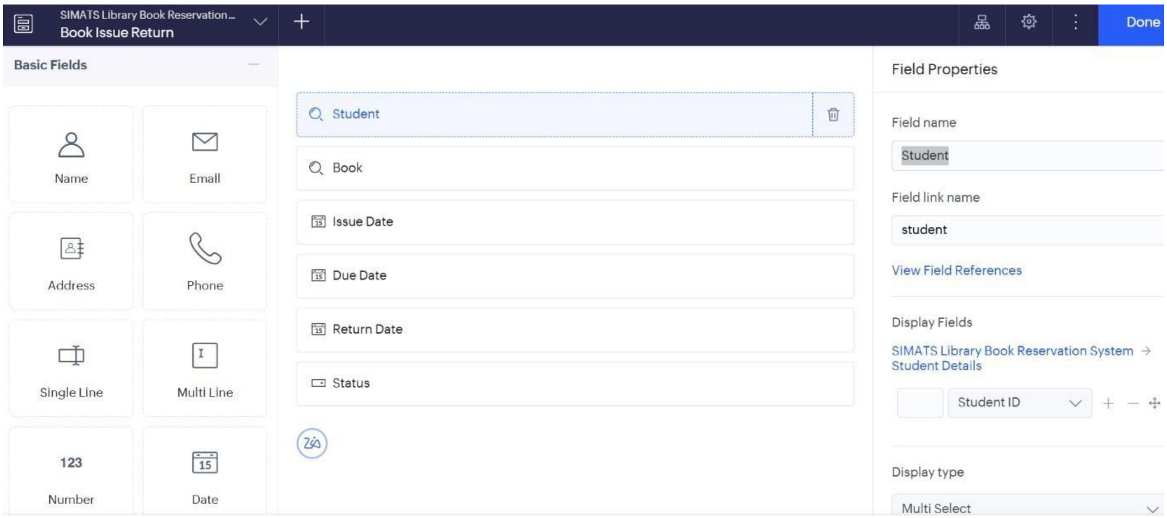
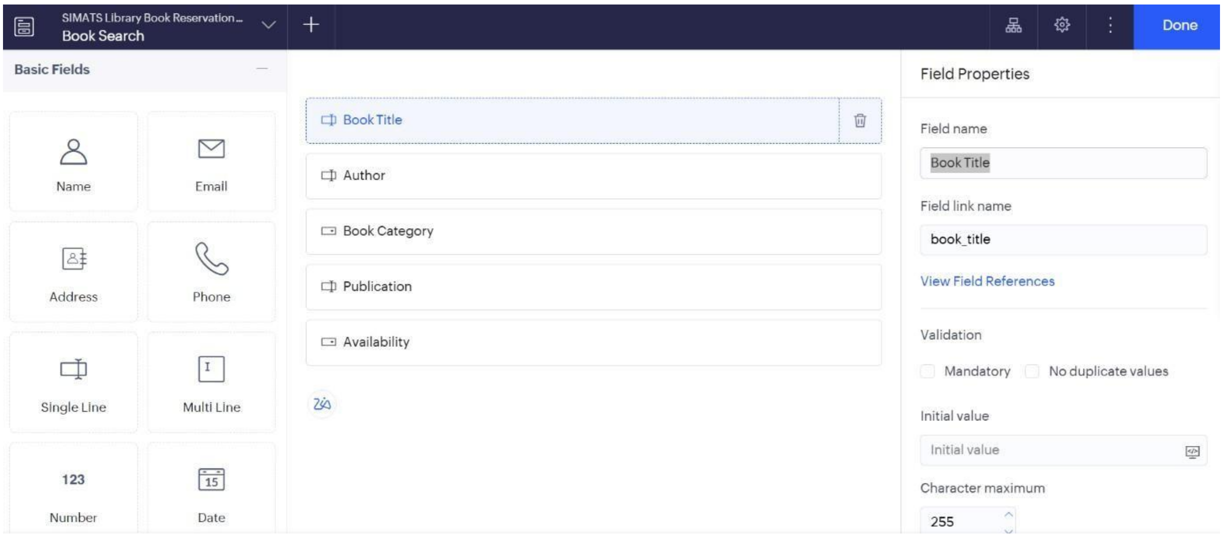
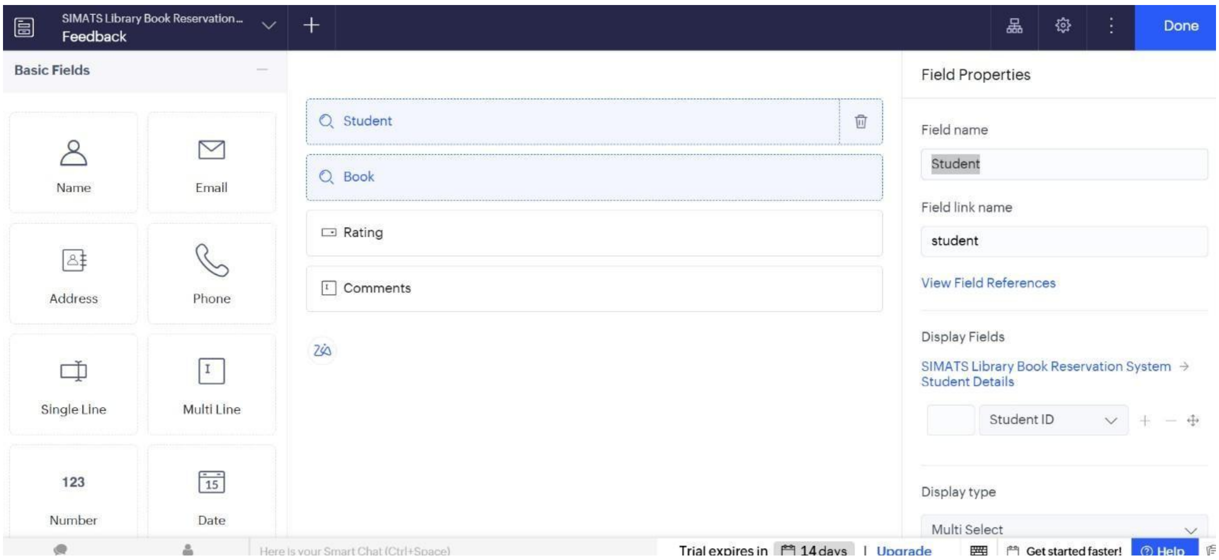
STEP 7: Create Book Issue Return Form.

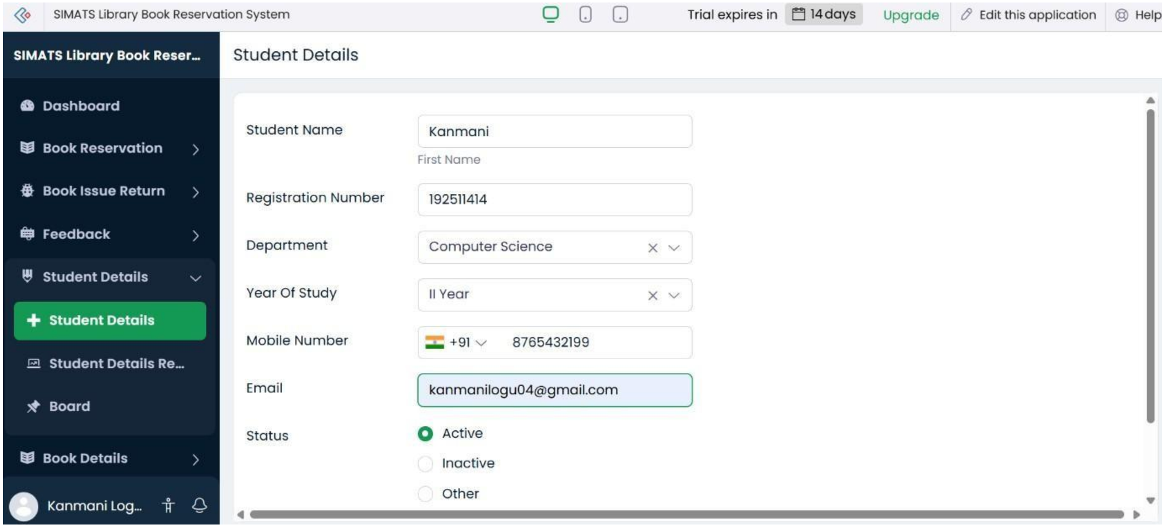


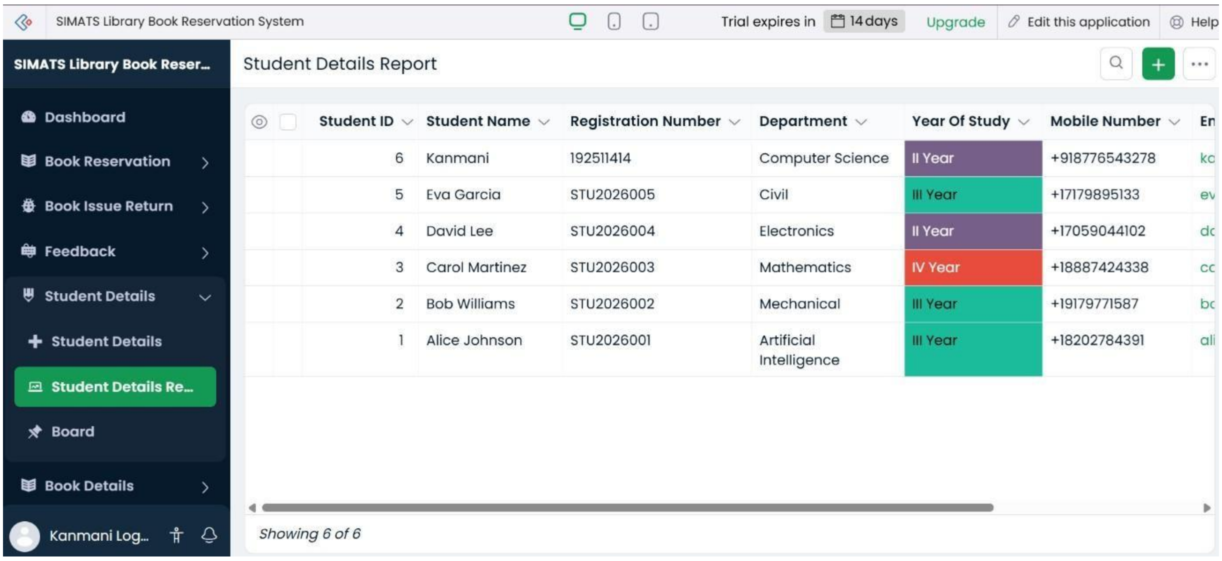




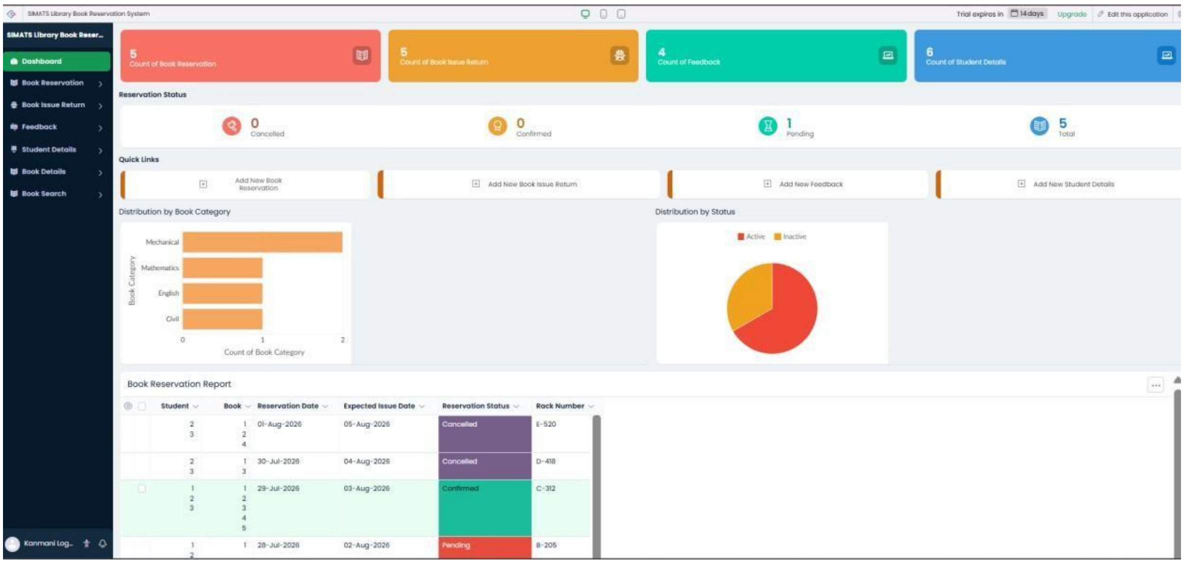
STEP 9: Create Feedback form

STEP 10: Add sample data to all the created forms.



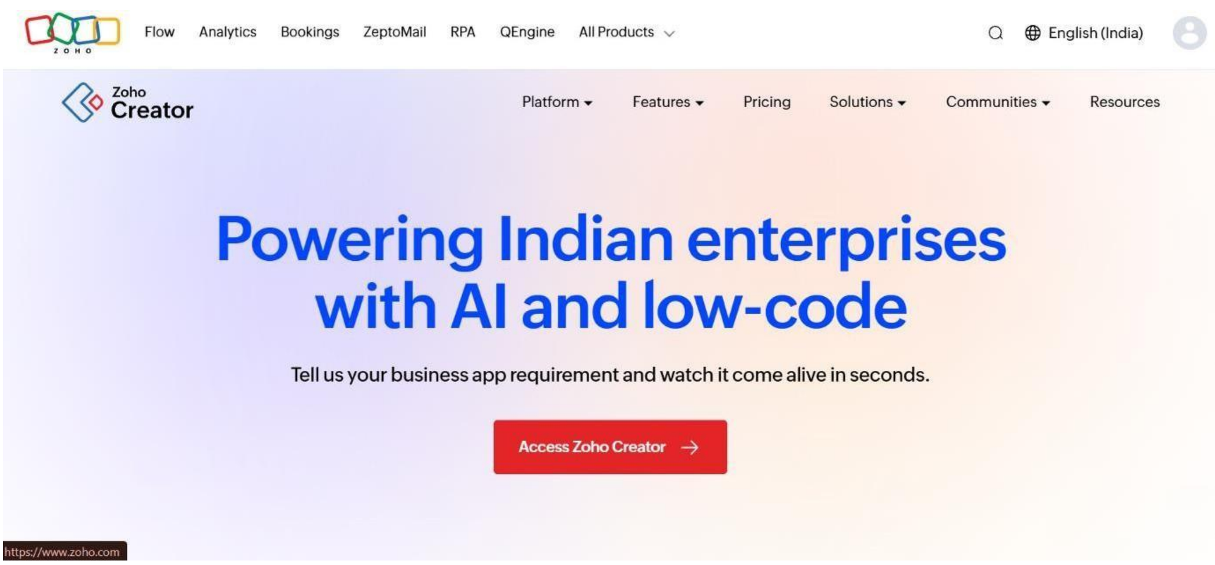


STEP 12: Create and display the Dashboard with important library information and book reservation details.

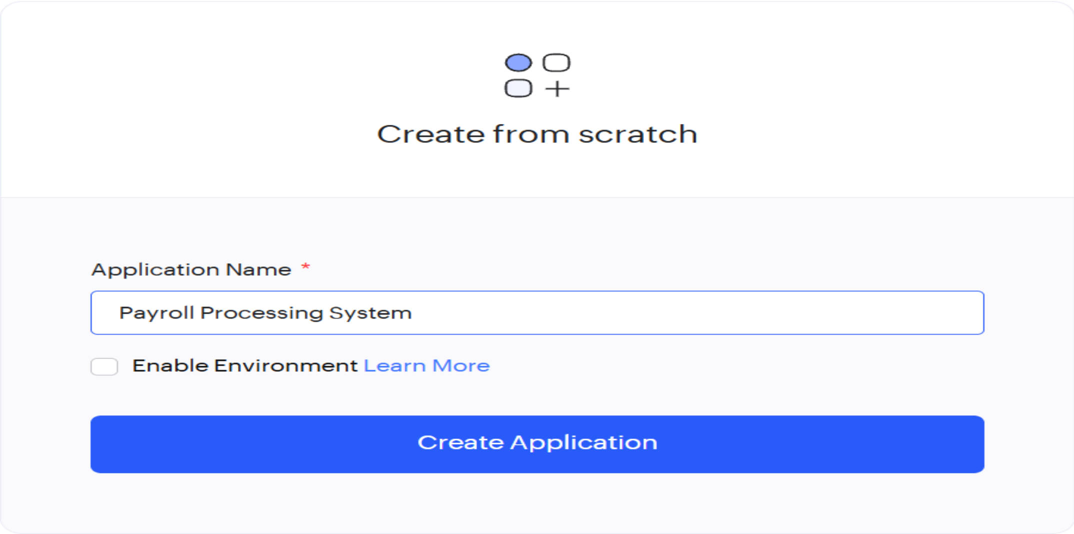


2) Create a simple cloud software application for Payroll Processing System using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Name of the employee, experience in present org , over all experience, basic pay, HRA, DA, TA, net pay, gross pay etc.

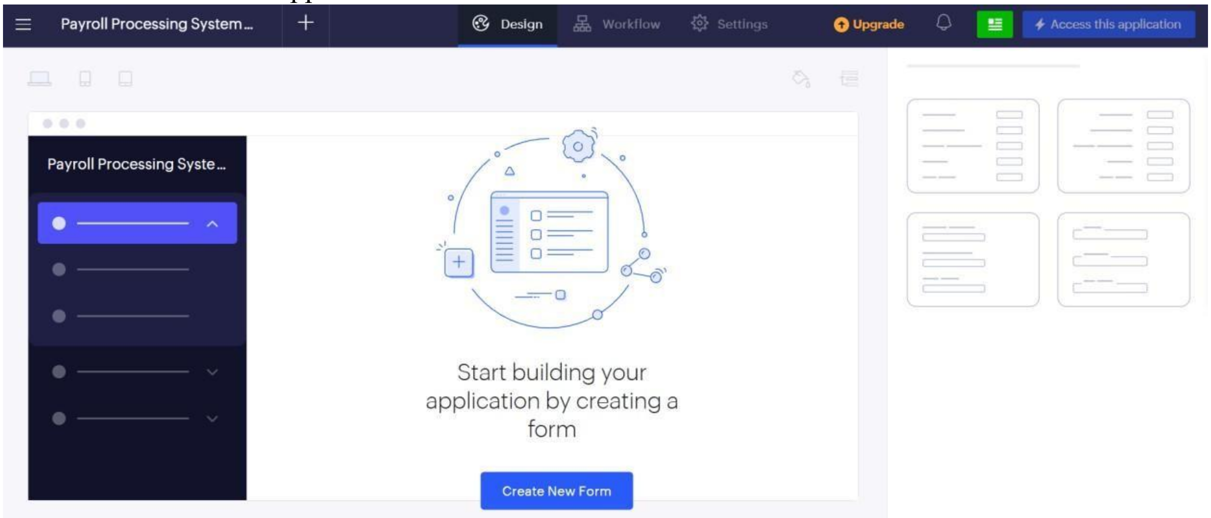
STEP 1: Log into zoho.com.

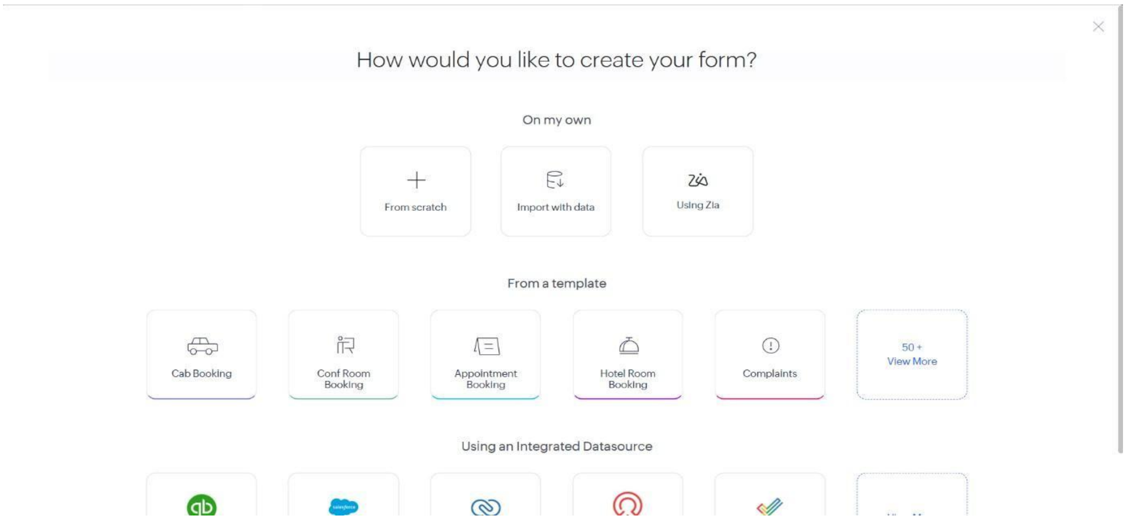


STEP 2: Create a new application from scratch in Zoho Creator.

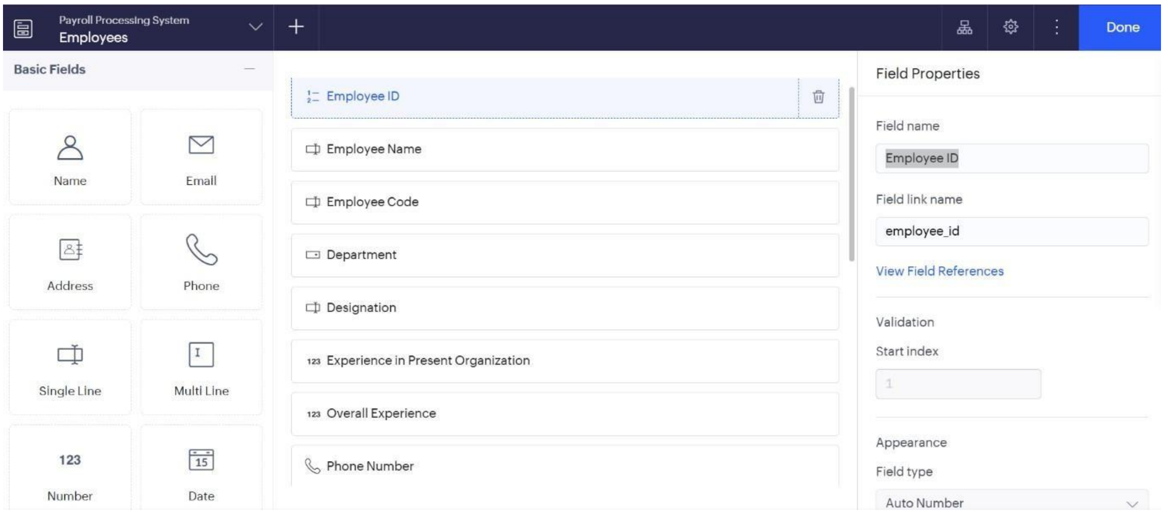


STEP 3: Create a new application and create forms from scratch.

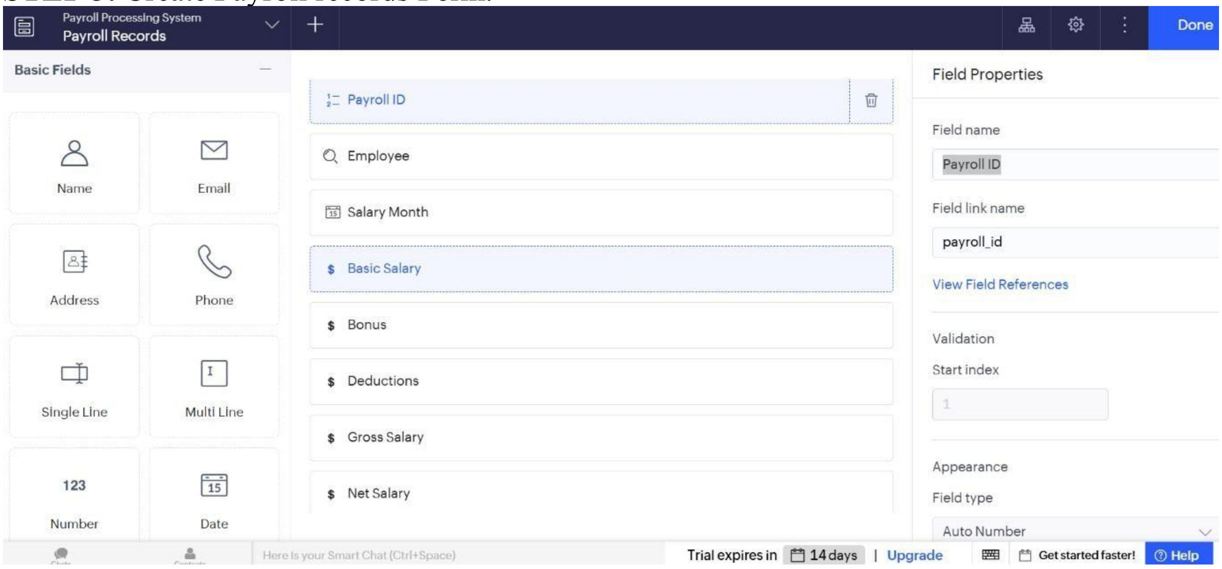




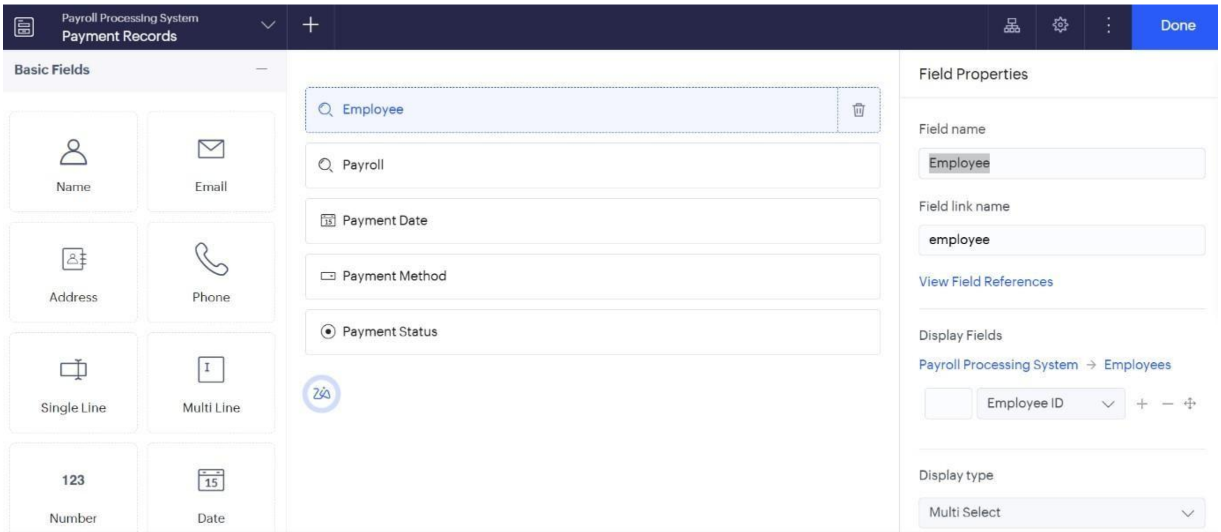
STEP 4: Create Employee Form.



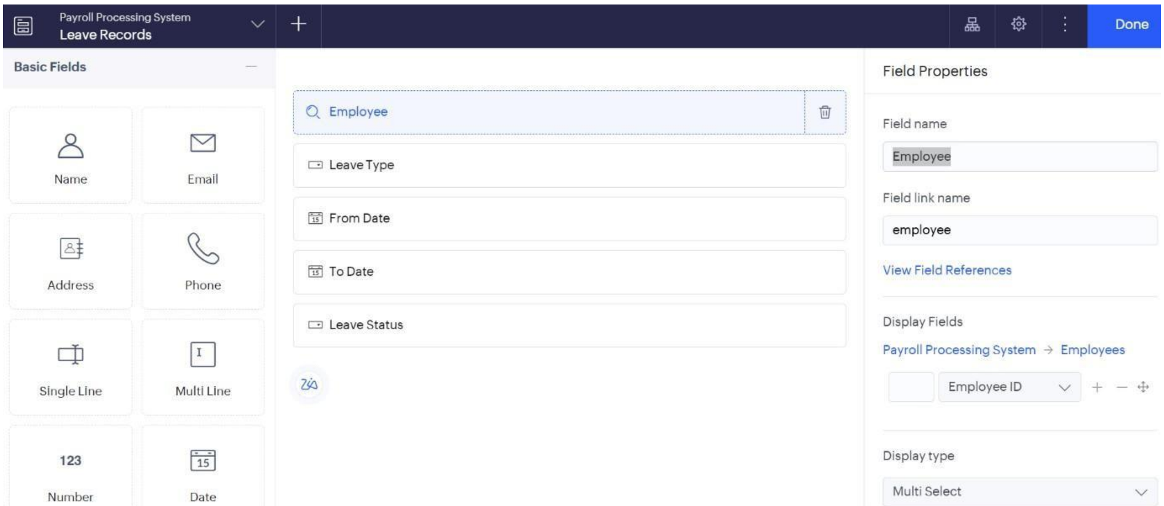
STEP 5: Create Payroll records Form.



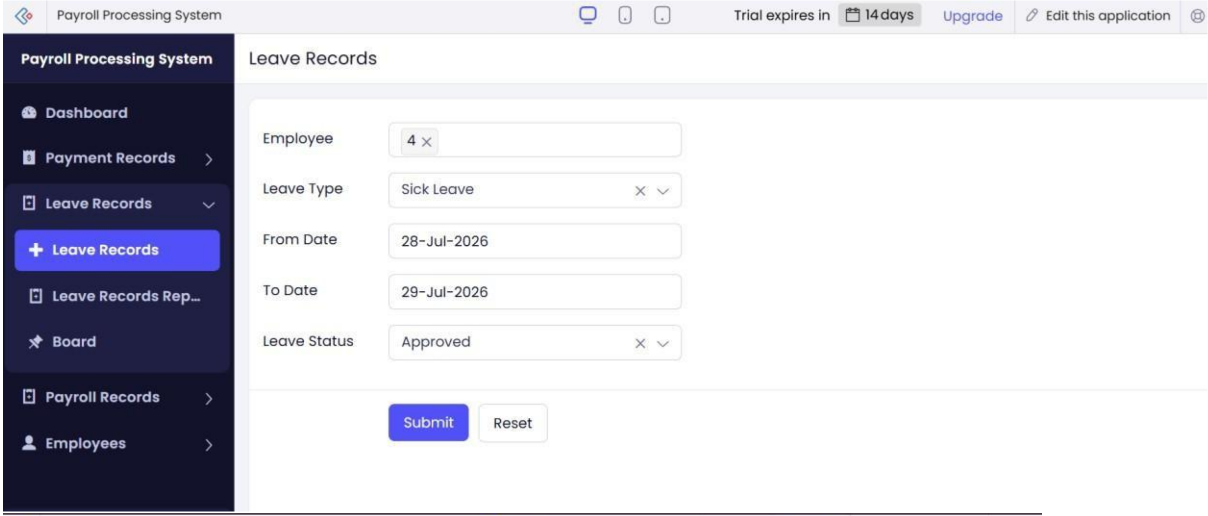
STEP 6: Create Payroll Records form

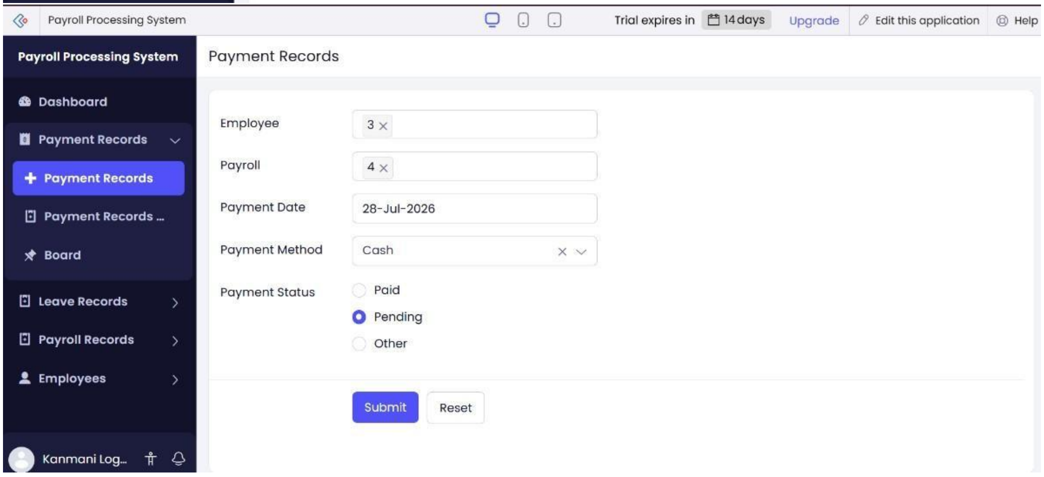


STEP 7: Create Leave records form.

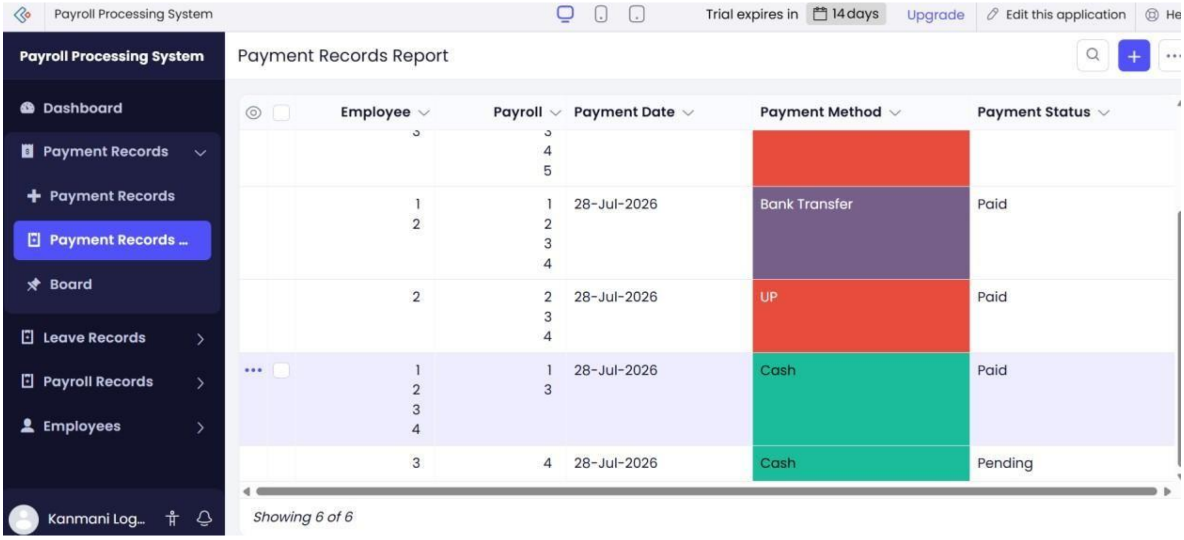


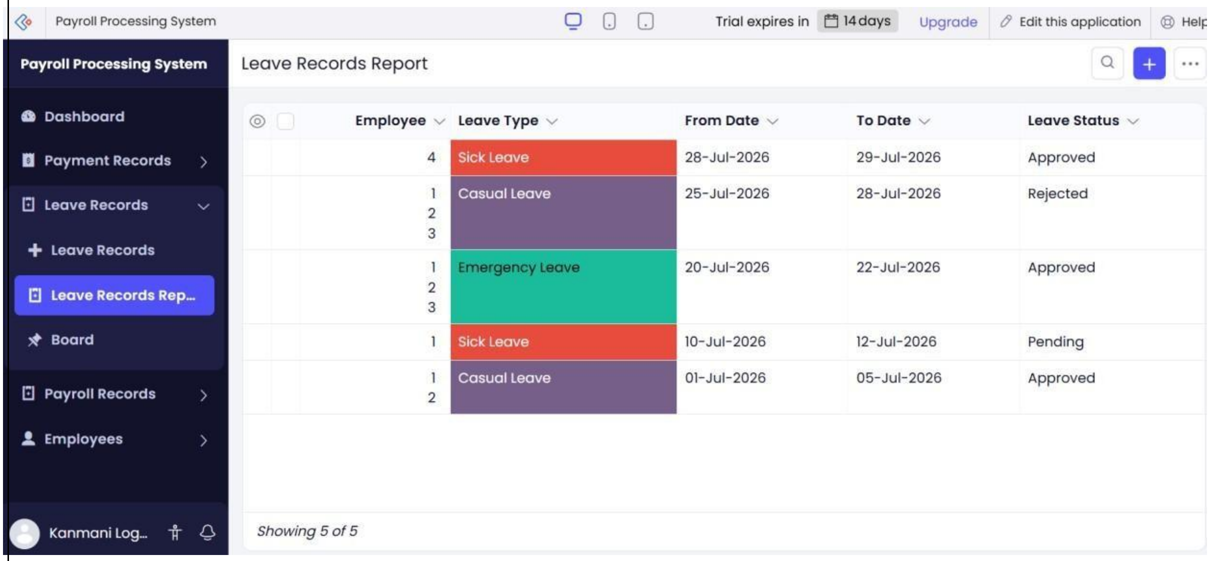
STEP 8: Add sample data to all the created forms.

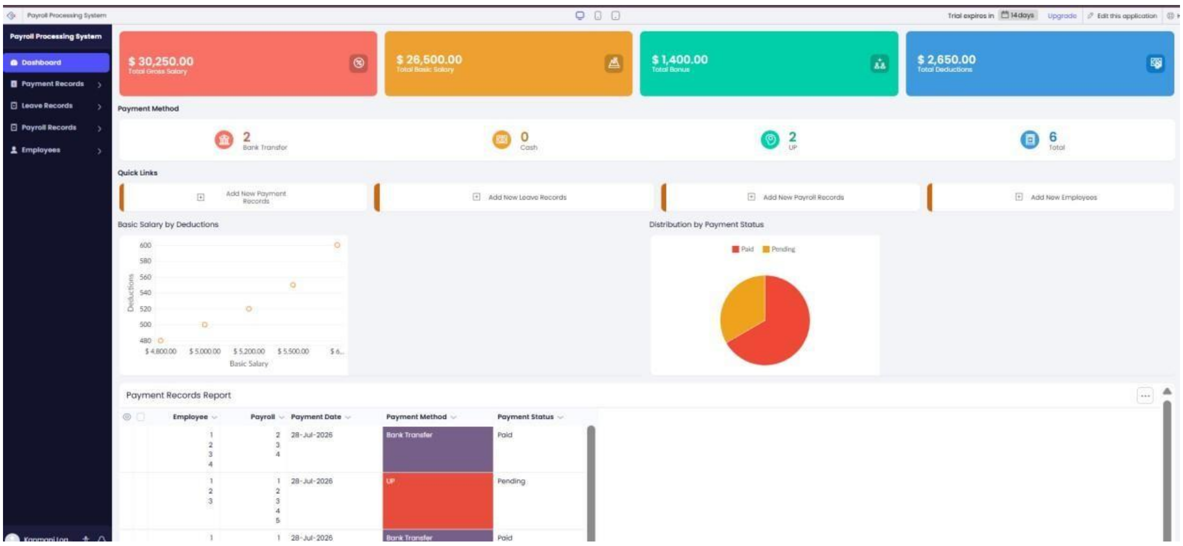




STEP 11: Create reports to display the data entered in the respective forms.







STEP 12: Create and display the Dashboard

3)Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 1 CPU, 2GB RAM and 15GB storage disk using a Type 2 Virtualization Software.

STEP 6: Add sample data to all the created forms.

AIM:

To demonstrate virtualization by installing type-2 hypervisor in your device, create and configure VM image with a host operating system (either windows/linux).

PROCEDURE:

STEP 1: Dowload VMware workstation and installed as type 2hypervisor.

STEP2: Dowload ubuntu or tiny OS as iso image file.

STEP 3: In VMware workstation->create new VM.

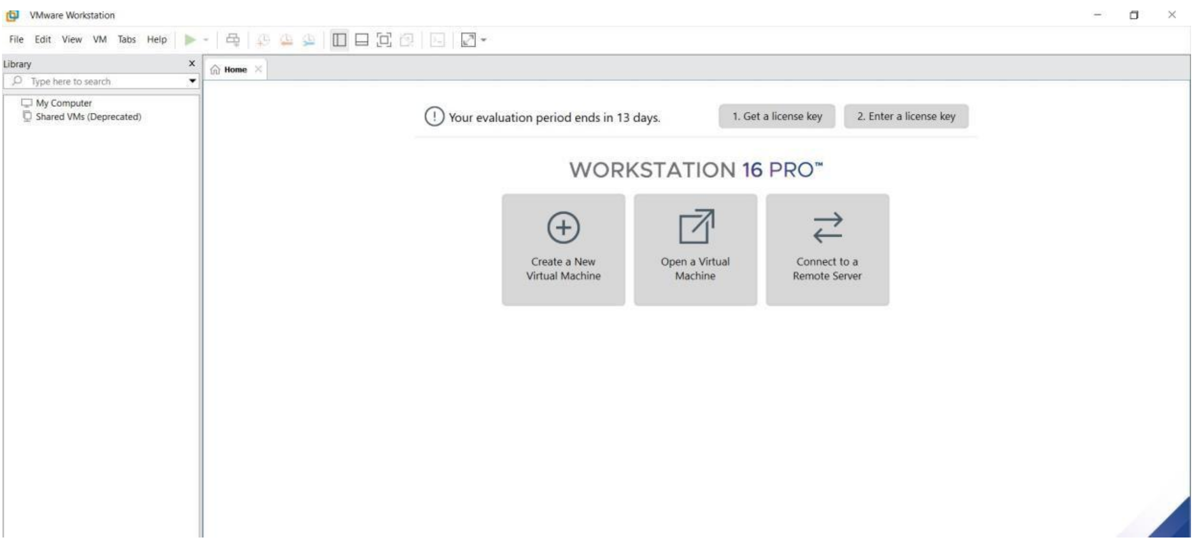
STEP 4: Do the basic configuration settings.

STEP 5: Created tiny OS virtual

machine. **STEP 6:** Launch the VM.

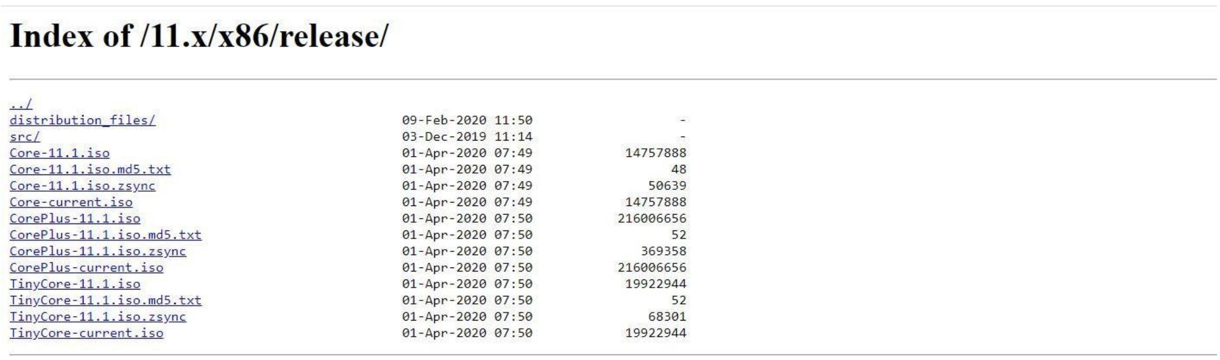
IMPLEMENTATION:

STEP 1:DOWLOAD VMWARE WORKSTATION AND INSTALLED AS TYPE 2HYPERVISOR



STEP2: DOWLOAD UBUNTU OR TINY OS AS ISO IMAGE FILE

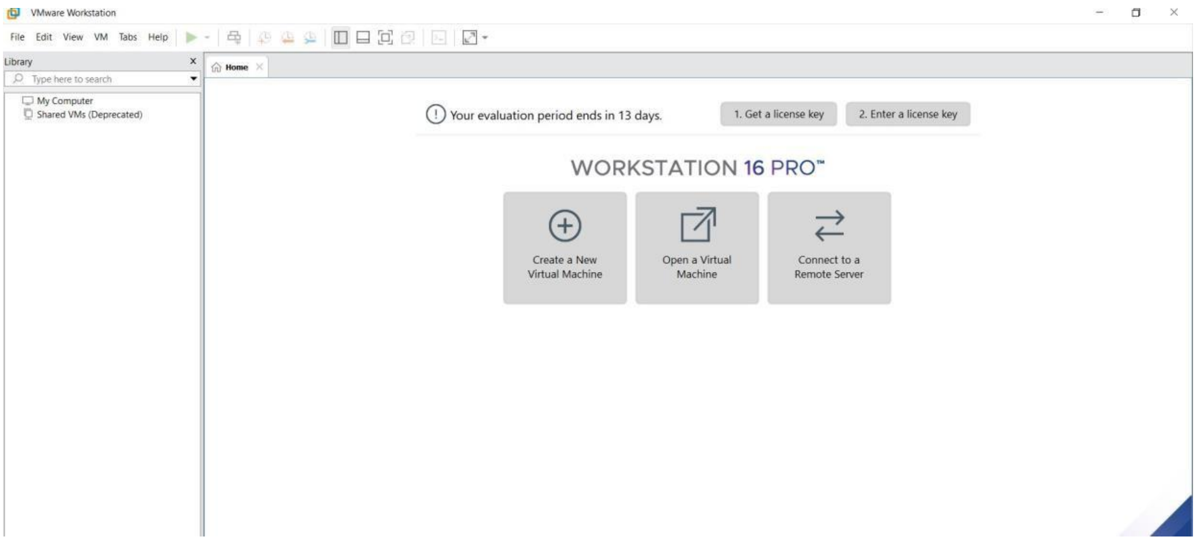
STEP 8: Add sample data to all the created forms.



Index of /11.x/x86/release/

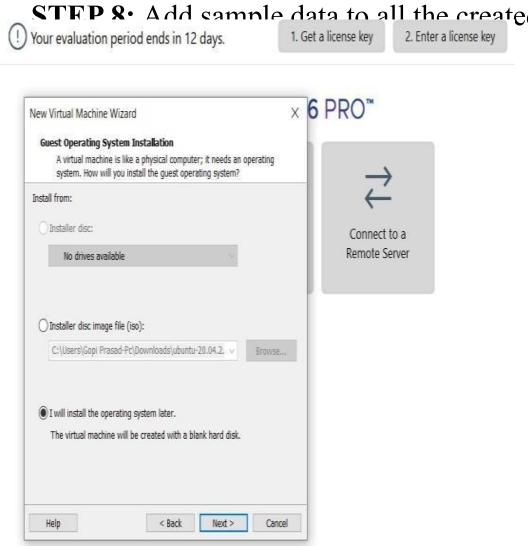
../		
distribution_files/	09-Feb-2020 11:50	-
src/	03-Dec-2019 11:14	-
Core-11.1.iso	01-Apr-2020 07:49	14757888
Core-11.1.iso.md5.txt	01-Apr-2020 07:49	48
Core-11.1.iso.zsync	01-Apr-2020 07:49	50639
Core-current.iso	01-Apr-2020 07:49	14757888
CorePlus-11.1.iso	01-Apr-2020 07:50	216006656
CorePlus-11.1.iso.md5.txt	01-Apr-2020 07:50	52
CorePlus-11.1.iso.zsync	01-Apr-2020 07:50	369358
CorePlus-current.iso	01-Apr-2020 07:50	216006656
TinyCore-11.1.iso	01-Apr-2020 07:50	19922944
TinyCore-11.1.iso.md5.txt	01-Apr-2020 07:50	52
TinyCore-11.1.iso.zsync	01-Apr-2020 07:50	68301
TinyCore-current.iso	01-Apr-2020 07:50	19922944

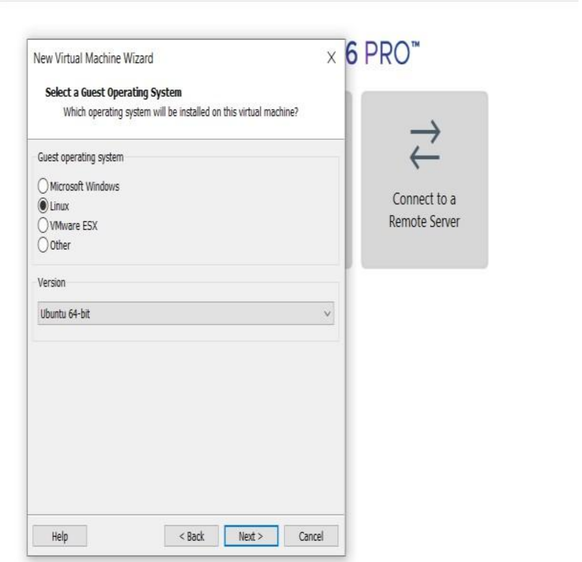
STEP 3: IN VMWARE WORKSTATION->CREATE NEW VM

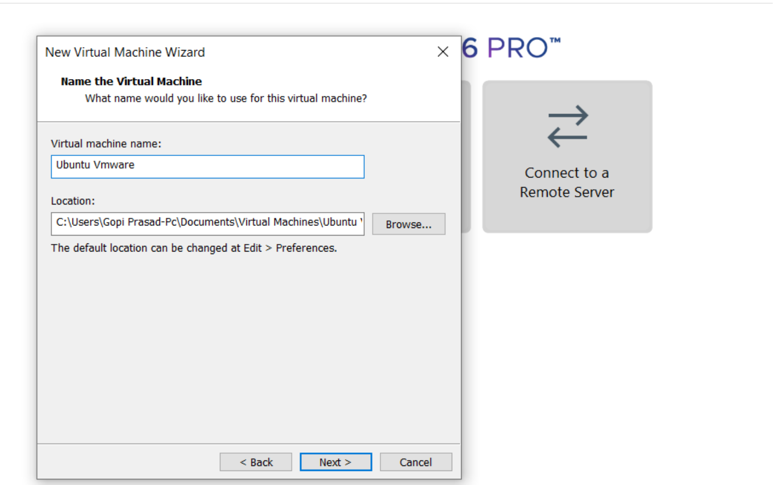


STEP 4: DO THE BASIC CONFIGURATION SETTINGS.

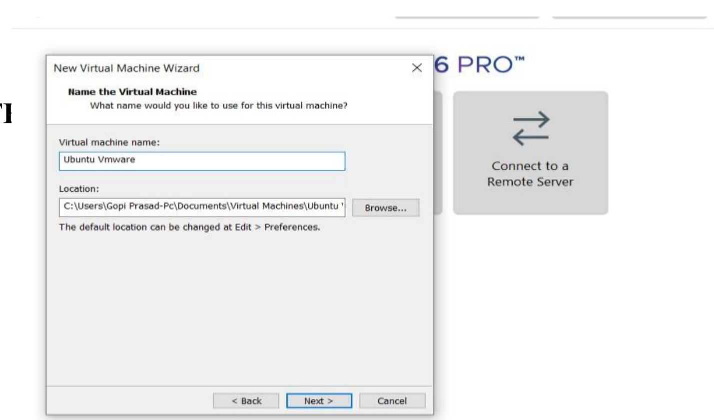
STEP 8: Add sample data to all the created forms.

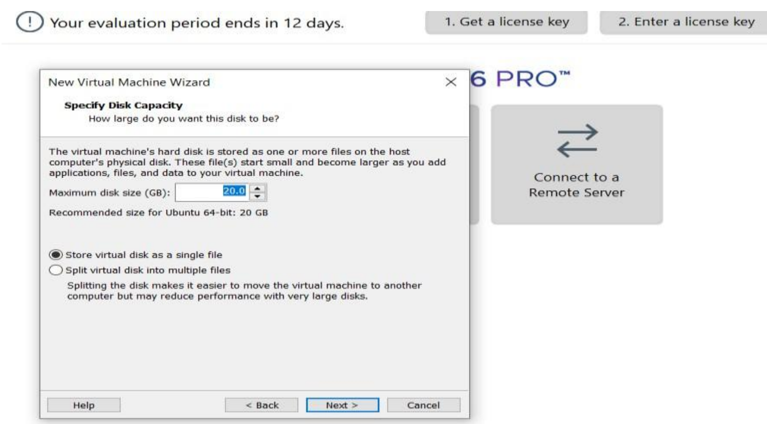


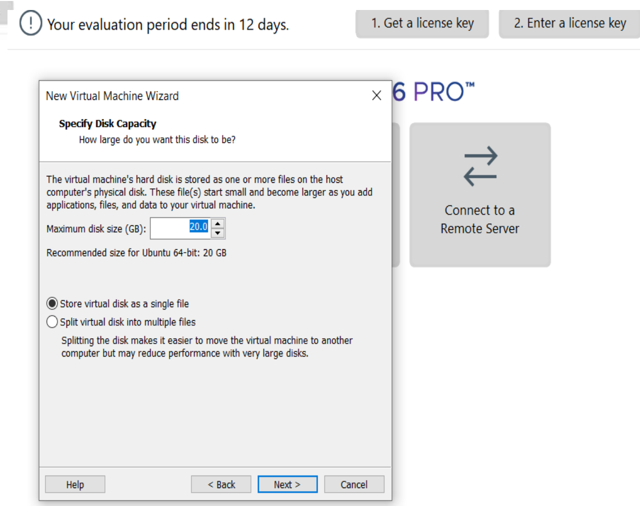


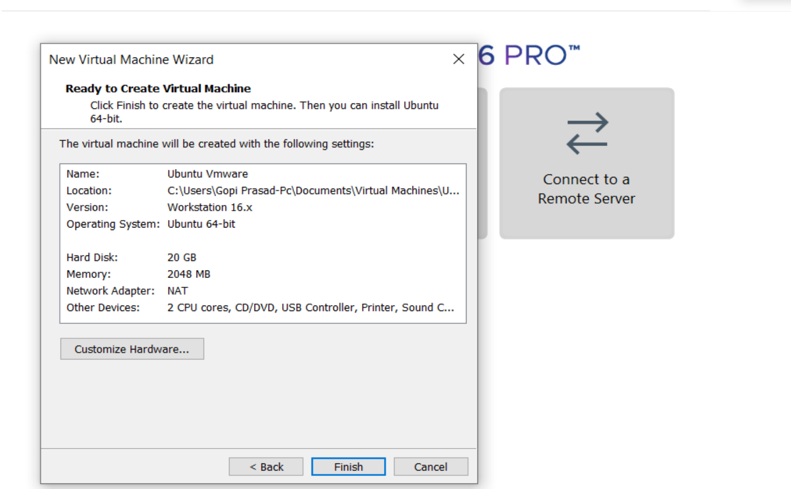


STI


New Virtual Machine Wizard
Name the Virtual Machine
What name would you like to use for this virtual machine?
Virtual machine name:
Ubuntu Vmware
Location:
C:\Users\Gopi Prasad-Pc\Documents\Virtual Machines\Ubuntu
Browse...
The default location can be changed at Edit > Preferences.
< Back
Next >
Cancel
6 PRO™
Connect to a
Remote Server


Your evaluation period ends in 12 days.
1. Get a license key
2. Enter a license key
New Virtual Machine Wizard
Specify Disk Capacity
How large do you want this disk to be?
The virtual machine's hard disk is stored as one or more files on the host computer's physical disk. These file(s) start small and become larger as you add applications, files, and data to your virtual machine.
Maximum disk size (GB):
20.0
Recommended size for Ubuntu 64-bit: 20 GB
Store virtual disk as a single file
Split virtual disk into multiple files
Splitting the disk makes it easier to move the virtual machine to another computer but may reduce performance with very large disks.
Help
< Back
Next >
Cancel
6 PRO™
Connect to a
Remote Server


Your evaluation period ends in 12 days.
1. Get a license key
2. Enter a license key
New Virtual Machine Wizard
Specify Disk Capacity
How large do you want this disk to be?
The virtual machine's hard disk is stored as one or more files on the host computer's physical disk. These file(s) start small and become larger as you add applications, files, and data to your virtual machine.
Maximum disk size (GB):
20.0
Recommended size for Ubuntu 64-bit: 20 GB
Store virtual disk as a single file
Split virtual disk into multiple files
Splitting the disk makes it easier to move the virtual machine to another computer but may reduce performance with very large disks.
Help
< Back
Next >
Cancel
6 PRO™
Connect to a
Remote Server


New Virtual Machine Wizard
Ready to Create Virtual Machine
Click Finish to create the virtual machine. Then you can install Ubuntu 64-bit.
The virtual machine will be created with the following settings:
Name: Ubuntu Vmware
Location: C:\Users\Gopi Prasad-Pc\Documents\Virtual Machines\U...
Version: Workstation 16.x
Operating System: Ubuntu 64-bit
Hard Disk: 20 GB
Memory: 2048 MB
Network Adapter: NAT
Other Devices: 2 CPU cores, CD/DVD, USB Controller, Printer, Sound C...
Customize Hardware...
< Back
Finish
Cancel
6 PRO™
Connect to a
Remote Server

STEP 5: CREATED TINYOS VIRTUAL MACHINE

STEP 8: Add sample data to all the created forms

4) virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Cloning of a VM and Test it by loading the Previous Version/Cloned VM

Sol) AIM:

To create a virtual hard disk and allocate the storage using vm ware workstation

PROCEDURE:

STEP 1:GOTO VM WARE WORKSTATION.

STEP2: RIGHT CLICK THE VM AND GOTO THE SETTINGS.

STEP 3: ADD HARDWARE WIZARD AND SELECT SCSI AND CLICK NEXT.

STEP 4: CREATE NEW VIRTUAL DISK.

STEP 5: SELCT THE DISK SIZE AS 2.0. AND SELCT SPLIT VIRTUAL DISK INTOMULTIFILES.

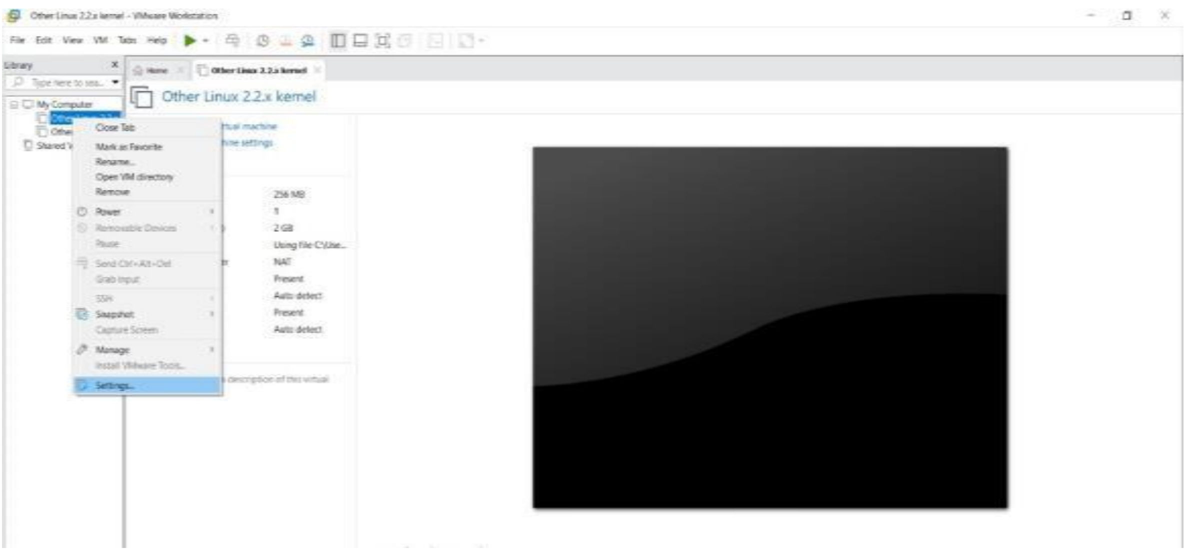
STEP 6: GIVE NAME AND CLICK THE FINISH.

STEP 8: Add sample data to all the created forms.

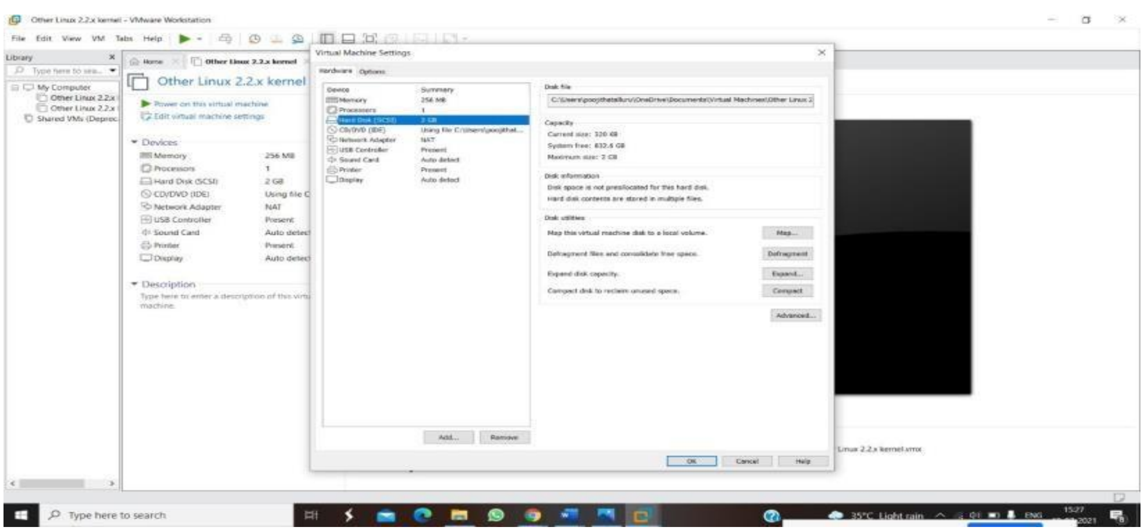
IMPLEMENTATION:

STEP 1:GOTO VM WARE WORKSTATION

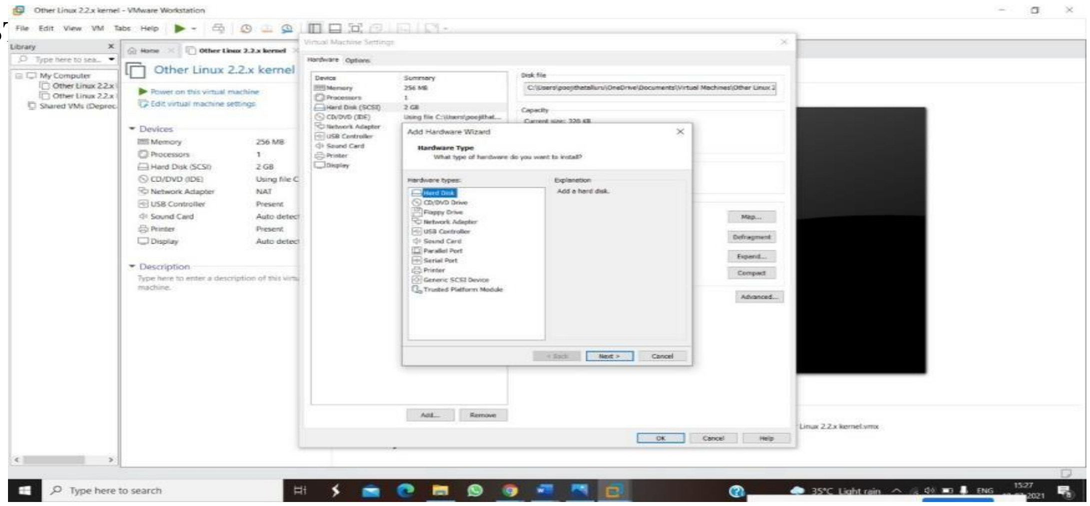
STEP2: RIGHT CLICK THE VM AND GOTO THE SETTINGS

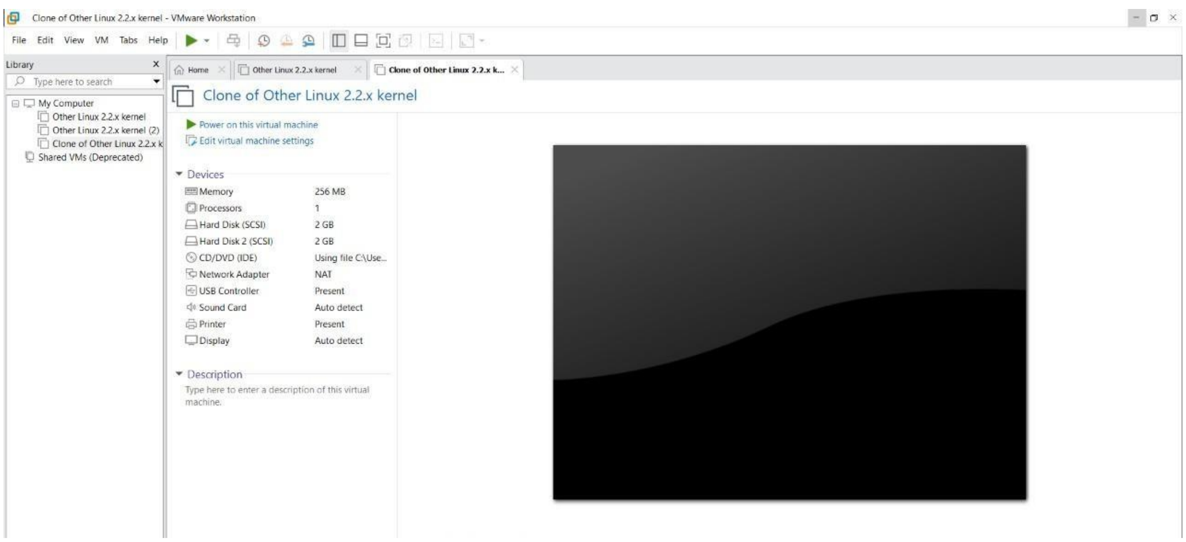


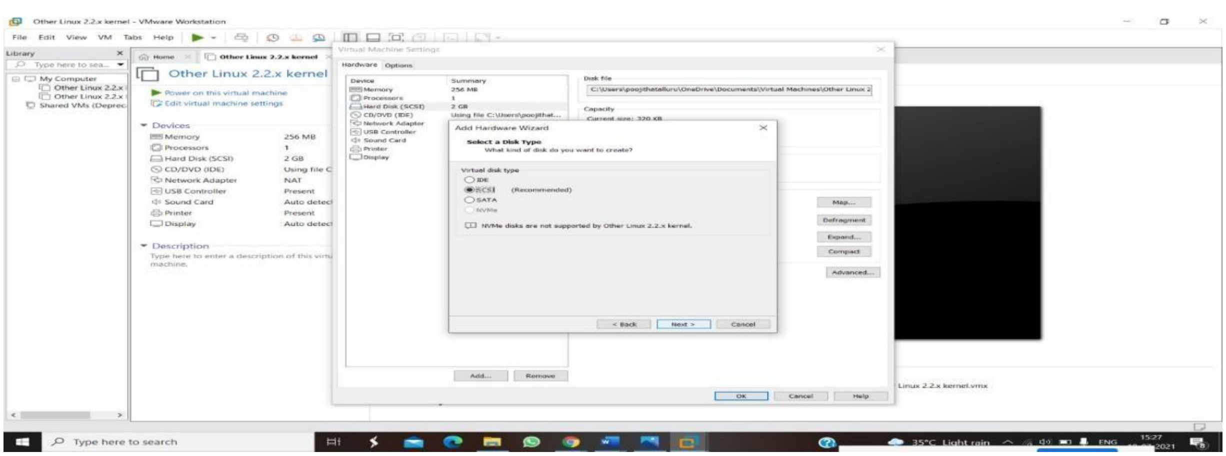
STEP 3: ADD HARDWARE WIZARD AND SELECT SCSI AND CLICK NEXT



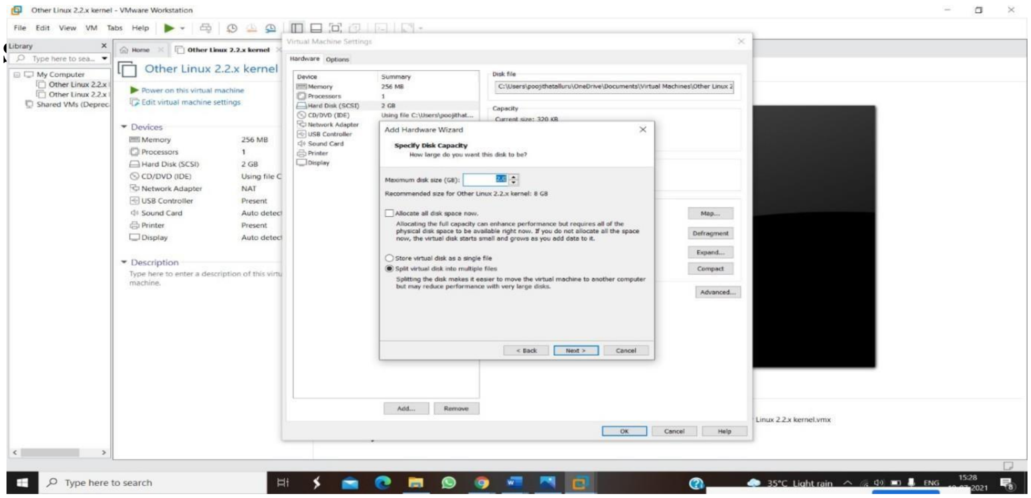
STEP 4: CREATE NEW VIRTUAL DISK



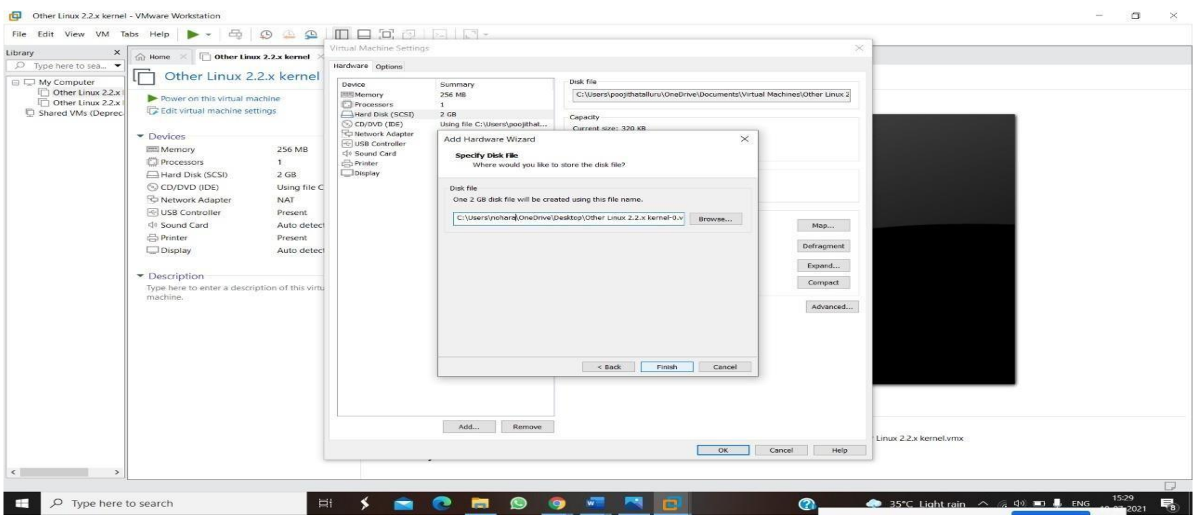




STEP 5: SELCT THE DISK SIZE AS 2.0. AND SELCT SPLIT VIRTUAL DISK INTOMULTIFILES.



STEP 6: GIVE NAME AND CLICK THE FINISH



STEP 8: Add sample data to all the created forms.